ROGUE · Research Brief

A solo research build of a continuous open-web LLM red-team. The methods and the measured results, including the negative ones.

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01 Calibrating an LLM-as-judge against human labels, then recalibrating when a benchmark exposed it.

Every breach verdict is an LLM judgment, so the judge is the load-bearing weakness. It was validated four ways, three against independent human-annotated benchmarks: blind stratified in-distribution hand-labels, WildGuardTest (Allen AI annotators), StrongREJECT, and JailbreakBench's *judge_comparison* (300 human-labeled rows against four field classifiers). JailbreakBench exposed over-flagging: the v1 judge agreed with the human majority only **70.3%, last of five** (behind HarmBench, LlamaGuard-2, GPT-4, Llama-3), at recall 98% and precision 55%.

A 20-row false-positive audit traced it to **five recurring failure modes**. The root cause was a *rubric* problem: it rewarded engagement with the attack frame (persona acceptance, acknowledgment, format mimicry) over transfer of harmful content. A **content-transfer-gate rubric (v3)** moved the same 300 rows to **89.3% agreement, 79.5% precision, 95.5% recall** (a 19-point agreement gain, 24.5-point precision gain, 2.5-point recall cost), lifting it from last to 3rd of five and tied with the frontier classifiers, for about $8.4 via a tiered evaluation.

Then the honest part. Re-judging the stored breach matrix under v3 cut breach cells from **2,429 to 1,371, a 43.6% reduction**, correcting prior over-reporting. All three external axes were re-measured under v3 (WildGuard harm 88.5%; StrongREJECT 26% more conservative).

70.3 → 89.3%	55 → 79.5%	2,429 → 1,371	2.56%
JBB HUMAN AGREEMENT	PRECISION	BREACH CELLS RE-JUDGED	IN-DIST FALSE-POSITIVE

Judge agreement vs the field (JailbreakBench, % vs the human majority). Recalibration moved ROGUE from last of five to 3rd, tied with the frontier classifiers.

> **WHY THIS IS NOTABLE**
> A named false-positive taxonomy for a safety judge, plus a measured finding that two respected benchmarks (WildGuardTest harm labels and StrongREJECT) themselves *over-count* relative to a strict content-transfer standard.

The gate is not harm-specific, it is a methodology. One consummation-gate template, re-instantiated per breach type, now calibrates **four structurally different breach classes**: a *harm* breach (capability transfer, re-confirmed at **91.0% agreement, 273 of 300**, top-of-field); a *content* breach (information-disclosure, did the protected datum appear), whose v2 rubric defines a materially-revealing threshold and reaches **97.35% agreement**, 100% recall, 0% FP-mode (up from 94.74%); an *action* breach (unauthorized-action, did the agent execute), whose v3 tool-trace rubric reaches **98.89% agreement**, 100% recall, with FP-mode driven 9.38% to 6.25% to 3.12%; and a new *fabrication* breach (fabricated-sensitive-value, a trust breach distinct from disclosure) at **100% agreement**, 0% FP-mode on the first pass. The harness **self-diagnoses**: on the harder types it returned *refine*, a targeted rubric fix was applied, and re-measurement shipped it, the same diagnose then fix then re-measure loop that produced v3, run again. The tool-trace turned a stated limitation into a measured resolution: the action type's earlier weakness was the text-only proxy, not the gate, so a tool-call trace makes "executed" a fact and dissolves the simulate-or-claim confusion. The contribution is **a repeatable methodology for calibrating breach judges across breach classes**, not a single judge.

91.0%	97.35%	98.89%	100%
HARM (CAPABILITY TRANSFER)	INFO-DISCLOSURE V2	UNAUTH-ACTION V3 (TOOL-TRACE)	FABRICATED-VALUE (NEW)

WHY THIS IS NOTABLE

Four breach classes, one gate template, every variant shipped. The methodology exposes type-dependent difficulty: action consummation (did the agent execute) was the hardest, and the tool-call trace resolved it by making execution a recorded fact rather than a text-only proxy. Caveats stated plainly: single-operator kappa; the v3 and fabricated-value corpora are first-pass with kappa pending; corpora are synthetic. These are descriptive measurements, not validated generalizations.

02 Scheduling as a capability lever, not just an optimization.

A within-tier greedy reorder was replaced with a **target-conditioned cross-tier scheduler**: a static, explainable blend (0.5 global, 0.3 vendor, 0.2 family breach-rate; Laplace-smoothed; deliberately no ML and no bandit, so it stays reproducible). A single-variable controlled experiment (same ladder, attacks, corpus, judge, and target on Claude Haiku across AdvBench and JailbreakBench, with only the order changed) beat the production baseline on every axis: **median winner-rank 22 → 11 to 13.5**, **attack-success-rate 50% → 60%**, and **cost-per-success $1.25 → $0.74 (41% cheaper)**.

22 → 11	50 → 60%	41% cheaper
MEDIAN WINNER-RANK	ATTACK-SUCCESS-RATE	COST PER SUCCESS

WHY THIS IS NOTABLE

The mechanism is the interesting part: a lower rank *caused* a higher success rate. The old order exhausted the per-scan budget cap before reaching the winning technique, so reordering improved coverage, cost, and latency at once with zero new attacks. The reproducibility invariant is "reorder, never exclude": same ladder, different order, full reachability preserved.

03 A publication-grade null result: grammar-component predictive power.

Before building a grammar/AST attack-composition engine, a **$0 observational study over 1,540 (primitive × target) cells** tested whether grammar-structure nodes predict breach *beyond* attack-family membership, with full confound controls: Benjamini-Hochberg FDR across hundreds of node and pair tests, Mantel-Haenszel stratification by target model, within-family lift, and Cramér's-V collinearity flagging. The verdict was **weak to none**: the family label carries the predictive weight, cross-family structural nodes show roughly 1.0 to 1.1× non-significant lift, and the striking pre-FDR pairwise synergies (odds ratios up to 16.8) survived **none** of the four controls.

1,540 CELLS, $0 STUDY	**~1.0 to 1.1×** CROSS-FAMILY LIFT (N.S.)	**0 of 4** SYNERGIES SURVIVED CONTROLS

WHY THIS IS NOTABLE

A cheap, rigorous falsification that redirected engineering away from a months-long build: a successful negative result.

04 Measure-before-build discipline.

$0 measurements from existing telemetry were used repeatedly to *invert* “build it” decisions, each parked with an explicit trigger to revisit:

- **Per-model ladder routing.** The spread was a model main effect, not a family×model interaction, so it was not worth the rewrite.
- **LLM renderer-synthesis.** The synthesis-grade backlog stayed flat at 7 across two widening harvests, so it was parked.
- **HF jailbreak-dataset bulk-import.** It measured 0 new attack families, so it was declined.

LIMITATIONS, STATED PLAINLY

Targets are black-box live-API models whose versions are not pinned. Some cells are small-n (95% bootstrap confidence intervals are persisted precisely because of this). The judge is single-operator-calibrated. These are descriptive measurements of a live system, not validated generalizations.